



D427 Practice Test 2



1. The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Cost – positive decimal

Write a SQL Statement to add up all costs in the Book table.

2. The Patient table has the following columns:

first_name – varchar(20)

last_name – varchar(30)

birthdate – date

patient_id – int

Which column should be designated as the primary key for the Patient table?

- a. first_name
- b. last_name
- c. birthdate
- d. patient_id

3. The Exam table has the following columns:

exam_id – int

exam_date – date

exam_reason – varchar (100)

patient_id – int

Which column should be designated as the foreign key for the Exam table?

- a. exam_id
- b. exam_date
- c. exam_reason
- d. patient_id

4. Which data type represents numbers with fractional values:

- a. Varchar
- b. Integer
- c. Binary
- d. Decimal

5. Which of the following is a DDL (Data Definition Language) command?

- a. INSERT
- b. SELECT
- c. CREATE INDEX
- d. UPDATE

6. Which of the following is a DML(Data Manipulation Language) command?

- a. CREATE VIEW
- b. CREATE TABLE
- c. INSERT
- d. ALTER INDEX

7.

```
CREATE TABLE Exam (  
  exam_id      INT      NOT NULL AUTO_INCREMENT,  
  exam_date    DATE     NOT NULL,  
  exam_reason  VARCHAR (100),  
  patient_id   INT      NOT NULL,  
  PRIMARY KEY (exam_id),  
  FOREIGN KEY (patient_id) REFERENCES Patient (patient_id) ON DELETE CASCADE  
);
```

Patient

<u>PK patient_id</u>
first name
last name
birthdate

Exam

<u>PK exam_id</u>
exam date
exam reason
FK patient_id

Looking at the Patient and Exam tables above and the CREATE TABLE for the Exam table with foreign key reference statement above, what would happen to exams in the Exam table that are linked to a patient if that patient is deleted.

- a. Those exams would remain in the database.
- b. Those exams would be deleted also.
- c. The Patient ID for those exams would be changed to NULL.
- d. Nothing would happen.

8.

```
CREATE TABLE Exam (  
  exam_id      INT      NOT NULL AUTO_INCREMENT,  
  exam_date    DATE     NOT NULL,  
  exam_reason  VARCHAR (100),  
  patient_id   INT      NOT NULL,  
  PRIMARY KEY (exam_id),  
  FOREIGN KEY (patient_id) REFERENCES Patient (patient_id) ON DELETE RESTRICT  
);
```

Patient

<u>PK patient_id</u>
first name
last name
birthdate

Exam

<u>PK exam_id</u>
exam date
exam reason
FK patient_id

Looking at the Patient and Exam tables above and the CREATE TABLE for the Exam table with foreign key reference statement above, what would happen to exams in the Exam table that are linked to a patient if that patient is deleted.

- a. Those patients would remain in the database.
- b. Those patients would also be deleted.
- c. The Patient ID for those exams would be changed to NULL.
- d. The delete of the exam record would not be allowed.

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```
CREATE TABLE Exam (  
  exam_id      INT      NOT NULL AUTO_INCREMENT,  
  exam_date    DATE     NOT NULL,  
  exam_reason  VARCHAR (100),  
  patient_id   INT      NOT NULL,  
  PRIMARY KEY (exam_id),  
  FOREIGN KEY (patient_id) REFERENCES Patient (patient_id) ON DELETE SET TO NULL  
);
```

Patient

<u>PK patient_id</u>
first name
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Exam

<u>PK exam_id</u>
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Looking at the Patient and Exam tables above and the CREATE TABLE for the Exam table with foreign key reference statement above, what would happen to exams in the Exam table that are linked to a patient if that patient is deleted.

- a. Those patients would remain in the database.
- b. Those exam records would also be deleted.
- c. The patient ID for those exams would be changed to NULL.
- d. The delete of the exam record would not be allowed.

10.

Which of the following are true about a materialized view (Choose 2)?

- a. It is a base table.
- b. It is stored.
- c. It must be refreshed whenever the base table changes.
- d. The results are stored as a temporary table.

11.

The Student table will have the following columns:

StudentID—positive integer

FirstName—variable-length string with up to 50 characters

MiddleInitial—fixed-length string with 1 character

LastName—variable-length string with up to 50 characters

DateOfBirth—date

AccountBalance—positive decimal value representing a balance of up to \$24,999, with 2 digits for cents

Write a SQL statement to create the Student table.

Do not add any additional constraints to any column beyond what is stated.

12.

The Classification table has the following columns:

ClassificationCode—integer, primary key

ClassificationDescription—variable-length string

The Vehicle table should have the following columns:

Name—variable-length string, maximum 30 characters

ClassificationCode—integer

Write a SQL statement to create the Vehicle table. Designate the ClassificationCode column in the Vehicle table as a foreign key to the ClassificationCode column in the Classification table.

13.

The Vehicle table has the following columns:

ID—integer, primary key

Make—variable-length string

Model—variable-length string

Year—integer

A new column must be added to the Vehicle table:

Column name: EngineSize

Data type: A positive decimal value less than or equal to 7.9 with one decimal.

Write a SQL statement to add the EngineSize column to the Vehicle table.

14.

The Song table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create a view named MyMusic that contains the Title, Genre, and Year columns for all songs. Ensure your result set returns the columns in the order indicated.

15.

A database has a view named MyMusic.

Write a SQL statement to delete the view named MyMusic from the database.

16.

The Pet table has the following columns:

PetID – integer

Name - variable-length string

Breed - variable-length string

Birthdate - date

Write a SQL statement to modify the Pet table to make the PetID column the primary key.

17.

The Dog table has the following columns:

dogID - integer, primary key

name - variable-length string

breedID - integer

birthdate - date

The Breed table has the following columns:

breedID—integer

breedDescription—varchar

Releases—integer

Write a SQL statement to designate the breedID column in the Dog table as a foreign key to the breedID column in the Breed table.

18.

The Song table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create an index named `idx_year` on the Year column of the Song table.

19.

The Podcast table has the following columns:

podcastID—integer, primary key, auto_increment

Title—variable-length string

Genre – variable-length string

Speaker—variable-length string

Minutes—integer

The following data needs to be added to the Podcast table:

Title	Speaker	Genre	Minutes
Rock Painting	Kecia McDonald	No Data	25

Write a SQL statement to insert the indicated data into the Podcast table.

20.

The Podcast table has the following columns:

PodcastID—integer, primary key, auto_increment

Title—variable-length string

Speaker—variable-length string

Minutes—integer

Write a SQL statement to delete the row with the PodcastID value of 33 from the Podcast table.

21.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Cost – positive decimal

Write a SQL query that will **add \$10 (for charity) to the cost of any book** that costs more than \$100 dollars.

22. Which query illustrates performing an outer join of the Movie table with a different table?

- a. `SELECT M.Title, A.Name FROM Movie M, Actor A
WHERE M.ActorID = A.ActorID;`
- b. `SELECT M.Title, A.Name FROM Movie M, Actor A
LEFT JOIN M.ActorID = A.MovieID;`
- c. `SELECT M.Title, A.Name FROM Movie M RIGHT JOIN Actor A
ON M.ActorID = A.ActorID;`
- d. `SELECT M.Title, A.Actor FROM Movie M
INNER JOIN Actor A ON M.ActorID = A.ActorID`

23.

Assume there are two tables, A and B.

Which rows will always be included in the result set if Table A is inner joined with Table B?

- a. Only rows in Tables A and B that share the join condition.
- b. All rows in Table B
- c. All rows in Table A
- d. Only rows in Tables A and B that do not share the join condition.

24.

The database contains a table named Book.

Write a SQL query to return all data from the Book table without directly referencing any column names.

25.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to retrieve the Title and Genre values for all records in the Book table with a Year value of 2020. Ensure your result set returns the columns in the order indicated.

26.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to display all Title values in alphabetical order A–Z.

27.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to output the unique Genre values and the number of books with each genre value from the Book table as GenreCount. Sort the results by the Genre in alphabetical order A–Z. Ensure your result set returns the columns in the order indicated.

28. The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer

Write a SQL query to display both the Title(if available) and the TotalSales for all YearSales. Ensure your result set returns the columns in the order indicated.

29.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to return how many books have a Year value of 2019.

30. A(n) _____ is a query that is embedded (or nested) inside another query.

- a. alias b. operator
- c. subquery d. view

31. All changes to a table structure are made using the _____ command, followed by a keyword that produces the specific changes a user wants to make.

- a. ALTER TABLE
- b. UPDATE TABLE
- c. COMMIT TABLE
- d. DELETE TABLE

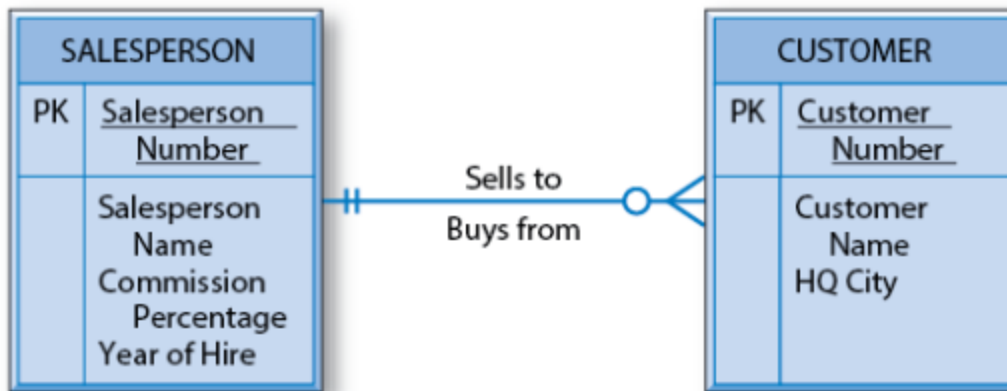
32. The SQL aggregate function that gives the number of rows containing non-null values for a given column is _____.

- a. COUNT b. MIN
- c. MAX d. SUM

33. The _____ condition of a JOIN is generally composed of an equality comparison between the foreign key and the primary key of related tables.

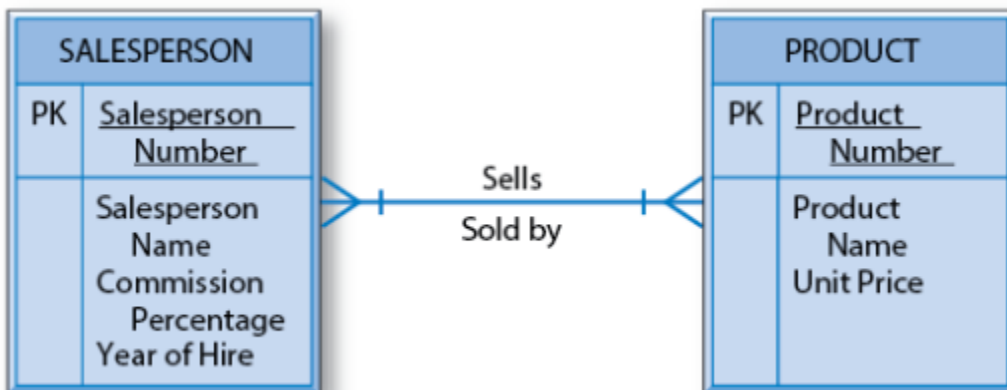
34. Which kind of relationship is displayed in the entity-relationship diagram below?

- a. Binary one-to-one
- b. Unary many-to-many
- c. Ternary one-to-one
- d. Binary one-to-many
- e. Unary one-to-one



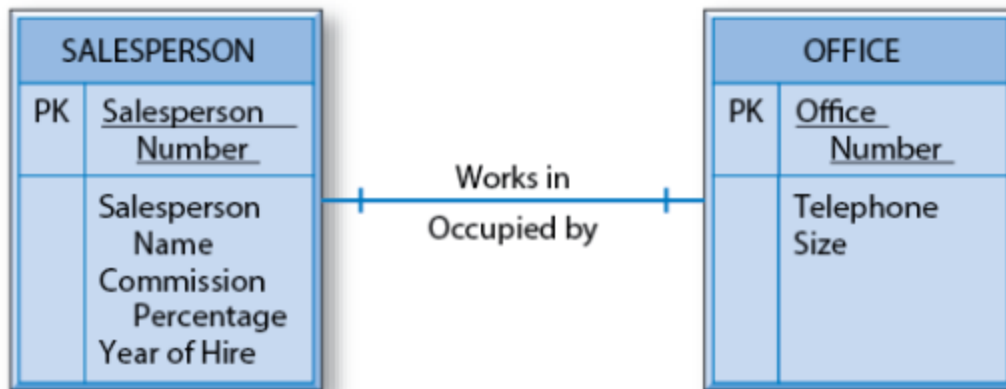
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- b. Unary many-to-many
- c. Ternary one-to-one
- d. Binary many-to-many
- e. Unary one-to-one



36. Which kind of relationship is displayed in the entity-relationship diagram below?

- a. Binary one-to-one
- b. Unary many-to-many
- c. Ternary one-to-one
- d. Binary many-to-many
- e. Unary one-to-one



37. Which data type can be designated to allow for storage of dates

- a. VARCHAR
- b. INTEGER
- c. DECIMAL
- d. TIMESTAMP/DATE

38.

Patient

patient_id	first_name	last_name	birthdate
101	Amy	Lin	1986-04-02
102	Viya	Prasad	1999-03-26
104	Blanca	Garcia	2000-12-11
107	Shelia	Brown	2004-11-11

Exam

exam_id	exam_date	exam_reason	patient_id
1	2023-04-04	Wellness check	104
2	2023-04-29	vaccine	107

Which query would produce a result set that listed all of the patients, regardless of whether they had an appointment in the Exam table or not.

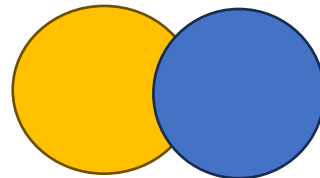
- a. `SELECT patient_id, first_name, last_name, exam_date
FROM Patient RIGHT JOIN Exam
ON Patient.patient_id = Exam.patient_id;`

- b. SELECT patient_id, first_name, last_name, birthdate, exam_date
FROM Patient LEFT JOIN Exam
ON Patient.patient_id = Exam.patient_id;
- c. SELECT patient_id, first_name, last_name, birthdate, exam_date
FROM Patient INNER JOIN Exam
ON Patient.patient_id = Exam.exam_id;
- d. SELECT patient_id, first_name, last_name, birthdate, exam_date
FROM Patient JOIN Exam
ON Patient.patient_id = Exam.patient_id;

39.

Models			
<u>model_id</u>	<u>lname</u>	<u>birthdate</u>	<u>model_type_id</u>
101	Almquist	9/1/05	CM
102	Chang	12/30/85	CP
103	Prasad	5/25/80	HF
104	Greene	5/23/83	PS

Model_Type	
<u>model_type_id</u>	<u>hourly_fee</u>
CM	500.00
HF	800.00
CP	800.00
PS	700.00
PM	700.00



Which query would produce a result set that listed all of the Model Types, regardless of whether a model was assigned to that type.

- a. SELECT model_id, lname, Models.model_type_id, hourly_fee
FROM Models
RIGHT JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;

- b.

```
SELECT model_id, Iname, Models.model_type_id, hourly_fee
FROM Models
LEFT JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;
```
- c.

```
SELECT model_id, Iname, Models.model_type_id, hourly_fee
FROM Models
INNER JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;
```
- d.

```
SELECT model_id, Iname, Models.model_type_id, hourly_fee
FROM Models
JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;
```

40.

Using Q39 tables, write a query to list all model IDs, the model type they are assigned and their hourly fees.

41.

Write a query to delete Amy Lin from the Customer table shown below.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip	Mobile Phone
101	Amy	Lin	123 A Street	Seattle	WA	98118	555-555-5555
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118	555-555-5556
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212	555-887-9264
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217	555-209-2784

42. Write a SQL statement to update Blanca Garcia's phone number in the Customer table below to 555-222-1234.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip	Mobile Phone
101	Amy	Lin	123 A Street	Seattle	WA	98118	555-555-5555
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118	555-555-5556
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212	555-887-9264
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217	555-209-2784

43. Write a SQL statement to retrieve all of the Customers from Seattle from the table below.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip	Mobile Phone
101	Amy	Lin	123 A Street	Seattle	WA	98118	555-555-5555
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118	555-555-5556
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212	555-887-9264
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217	555-209-2784

44. Write the CREATE TABLE statement to create the Exam table below with exam_id and patient_id as integers, and exam reason to have 250 characters of variable length. Make sure to designate the primary key and foreign key of the table.

Patient

PK patient_id

first name

last name

birthdate

Exam

PK exam_id

exam date

exam reason

FK patient_id

45. Write a query to pull all of the data from the Bagel table below. Order the result set alphabetically by Bagel ID.

Bagel ID	Bagel Name	Bagel Description	Cost/Bagel
RB	Radically Raisin Bagel	Raisin Bagel	2.00
EB	All the Stuff Bagel	Everything Bagel	2.35
CB	Chip off the Block Bagel	Chocolate Chip Bagel	2.25
MB	Keepin' it Healthy Bagel	Multigrain Bagel	2.00
NB	Classic New York Bagel	New York Bagel	1.75

46. Write a query to count the number of customers in each city.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip
101	Amy	Lin	123 A Street	Seattle	WA	98118
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217
108	Anna	Cruz	444 Apple Lane	Troutdale	OR	97708
110	Amy	Smith	39878 Acorn Place	Chicago	IL	60435

47. Write a query that will return all of the cities from the Customer table above. Do not include duplicates (list each city only once).

48. Write a query to list customers from Chicago or Seattle.

Customer ID	First Name	Last Name	Address	City	State	Zip
101	Amy	Lin	123 A Street	Seattle	WA	98118
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217
108	Anna	Cruz	444 Apple Lane	Troutdale	OR	97708
110	Amy	Smith	39878 Acorn Place	Chicago	IL	60435

49.

student_grades

studentID	lastName	grade
44444	Jones	B
24352	Chang	A
98724	Wu	B
98789	Chatham	A
98887	Fredricks	C
00878	Moore	A
22078	Prasad	C
79692	Perez	A

Refer to the given SELECT statement and the table above.

```
SELECT grade, COUNT(*) AS gradeCount  
FROM student_grades  
GROUP BY grade;
```

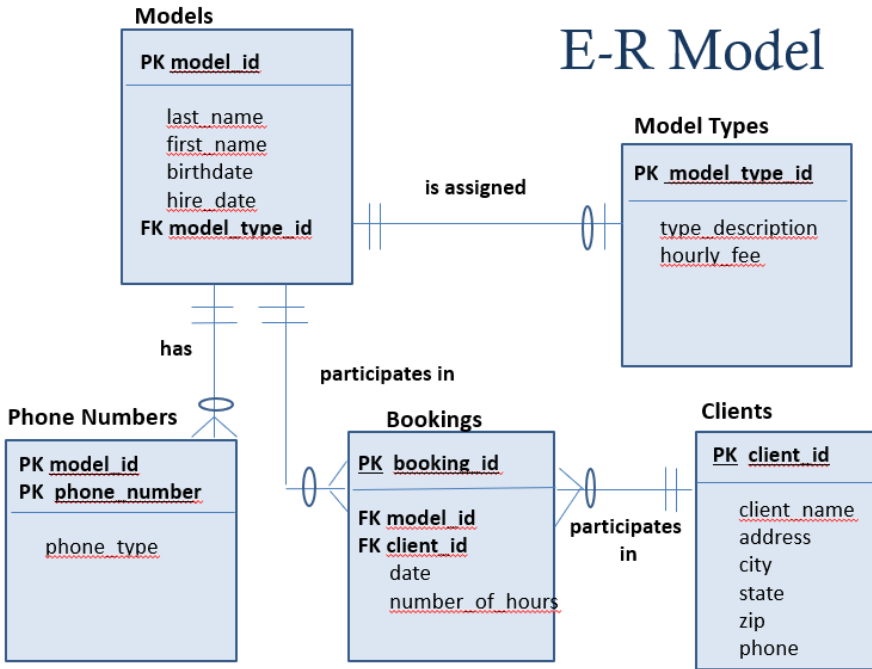
Which clause added to the statement finds grades with a count greater than 2?

Which clause added to the statement finds grades with a count greater than 2?

- a. WITH gradeCount <> DUPLICATE
- b. NOT gradeCount = NULL
- c. HAVING gradeCount > 2
- d. INCLUDING gradeCount > 2

50. Using the student_grades table listed above, write a query to count how many students have each grade (A, B, and C).

E-R Model



51. Using the Model_Types table in the E-R diagram above, write a query to find the highest model hourly fee.

52. Write a query to **change the hourly_fee to add \$100 more** for model types with an **hourly_fee less than \$600**.

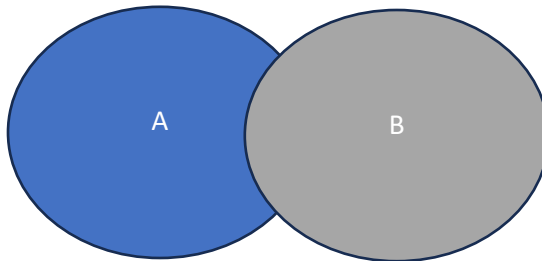
53.

Which query illustrates performing an outer join of the Movie table with a different table?

- SELECT M. Title, A.Actor FROM Movie M, Actor A
WHERE M.ActorID = A.ID
- SELECT M.Title, A.Actor FROM Movie M
LEFT JOIN Movie MB ON M.ID = MB.ID,_Actor A
- SELECT M. Title, A.Actor FROM Movie M
INNER JOIN Actor A ON M. ActorID = A.ID
- SELECT M. Title, A.Actor FROM Movie M
RIGHT JOIN Actor A ON M.ActorID = A.ID

54. Which rows will always be included in the result set if Table A is right joined with Table B?

- a. Only rows in Tables A and B that share the join condition
- b. All rows in Table B and any rows in Table A that share the join condition
- c. All rows in Table A and any rows in Table B that share the join condition
- d. Only rows in Tables A and B that do not share the join condition



55.

The Movie table has the following columns:

```
CREATE TABLE Movie (  
  MovieID INT NOT NULL,  
  MovieName VARCHAR(25) NOT NULL,  
  ReleaseDate DATE  
);
```

Canadians like long movie titles (Bill and Ted's Excellent Adventures). Modify the **MovieName** column in the Movie table so that the variable length string has a **maximum of 50 characters**.

56.

The Employee table has the following columns:

EmployeeID – integer, primary key
FirstName – variable-length character string
LastName – variable-length character string
EmployeeType – variable-length character string
Salary – decimal, positive

Write a SQL statement to pull the **EmployeeType and the lowest salary by EmployeeType** in the Employee table:

D427 Practice Test 2 – ANSWER KEY

Grade this test all or nothing like the OA does! No partial credit is given on the OA (even in SQL coding questions.) **You must score 17/25 on the OA to pass**, so try to score 90% on these practice tests & PreA, to be confident you will pass!

1. Write a SQL Statement to add up all costs in the Book table.

```
SELECT SUM(Cost)
FROM Book;
```

2. The Patient table has the following columns:

first_name – varchar(20)

last_name – varchar(30)

birthdate – date

patient_id – int

Which column should be designated as the primary key for the Patient table?

- a. first_name
- b. last_name
- c. birthdate
- d. patient_id

3. The Exam table has the following columns:

exam_id – int

exam_date – date

exam_reason – varchar (100)

patient_id – int

Which column should be designated as the foreign key for the Exam table?

- a. exam_id
- b. exam_date
- c. exam_reason
- d. patient_id

4. Which data type represents numbers with fractional values:

- a. Varchar
- b. Integer
- c. Binary
- d. Decimal

5. Which of the following is a DDL command?

- a. INSERT
- b. SELECT
- c. CREATE INDEX
- d. UPDATE

6. Which of the following is a DML command?

- a. CREATE VIEW
- b. CREATE TABLE
- c. INSERT
- d. ALTER INDEX

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```
CREATE TABLE Exam (  
  exam_id      INT      NOT NULL AUTO_INCREMENT,  
  exam_date    DATE     NOT NULL,  
  exam_reason  VARCHAR (100),  
  patient_id   INT      NOT NULL,  
  PRIMARY KEY (exam_id),  
  FOREIGN KEY (patient_id) REFERENCES Patient (patient_id) ON DELETE CASCADE  
);
```

Patient

<u>PK patient_id</u>
first name
last name
birthdate

Exam

<u>PK exam_id</u>
exam date
exam reason
FK patient_id

Looking at the Patient and Exam tables above and the CREATE TABLE for the Exam table with foreign key reference statement above, what would happen to exams in the Exam table that are linked to a patient if that patient is deleted.

- a. Those exams would remain in the database.
- b. Those exams would be deleted also.
- c. The Patient ID for those exams would be changed to NULL.
- d. Nothing would happen.

8.

```
CREATE TABLE Exam (  
  exam_id      INT      NOT NULL AUTO_INCREMENT,  
  exam_date    DATE     NOT NULL,  
  exam_reason  VARCHAR (100),  
  patient_id   INT      NOT NULL,  
  PRIMARY KEY (exam_id),  
  FOREIGN KEY (patient_id) REFERENCES Patient (patient_id) ON DELETE RESTRICT  
);
```

Patient

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first name
last name
birthdate

Exam

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exam date
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  exam_reason  VARCHAR (100),  
  patient_id   INT      NOT NULL,  
  PRIMARY KEY (exam_id),  
  FOREIGN KEY (patient_id) REFERENCES Patient (patient_id) ON DELETE SET TO NULL  
);
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Patient

PK patient_id
first name
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Looking at the Patient and Exam tables above and the CREATE TABLE for the Exam table with foreign key reference statement above, what would happen to exams in the Exam table that are linked to a patient if that patient is deleted.

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- c. The patient ID for those exams would be changed to NULL.
- d. The delete of the exam record would not be allowed.

10.

Which of the following are true about materialized view (Choose 2)?

- a. It is a base table.
- b. It is stored.
- c. It must be refreshed whenever the base table changes.
- d. The results are stored as a temporary table.

11.

The Student table will have the following columns:

StudentID—positive integer

FirstName—variable-length string with up to 50 characters

MiddleInitial—fixed-length string with 1 character

LastName—variable-length string with up to 50 characters

DateOfBirth—date

AccountBalance—positive decimal value representing a balance of up to \$24,999, with 2 digits for cents

Write a SQL statement to create the Student table.

Do not add any additional constraints to any column beyond what is stated.

CREATE TABLE Student (

StudentID INT UNSIGNED,

FirstName VARCHAR(50),

```

    MiddleInitial CHAR(1),
    LastName VARCHAR(50),
    DateOfBirth DATE,
    AccountBalance DECIMAL(7,2) UNSIGNED
    CHECK (AccountBalance < 25000) OR CHECK (AccountBalance <= 24999.99)
);

```

12.

The Classification table has the following columns:

ClassificationCode—integer, primary key

ClassificationDescription—variable-length string

The Vehicle table should have the following columns:

Name—variable-length string, maximum 30 characters

ClassificationCode—integer

Write a SQL statement to create the Vehicle table. Designate the ClassificationCode column in the Vehicle table as a foreign key to the ClassificationCode column in the Classification table.

```

CREATE TABLE Vehicle (
    Name VARCHAR(30),
    ClassificationCode INT,
    FOREIGN KEY (ClassificationCode) REFERENCES Classification(ClassificationCode)
);

```

13.

The Vehicle table has the following columns:

ID—integer, primary key

Make—variable-length string

Model—variable-length string

Year—integer

A new column must be added to the Vehicle table:

Column name: EngineSize

Data type: A positive decimal value less than or equal to 7.9 with one decimal place.

Write a SQL statement to add the EngineSize column to the Vehicle table.

```

ALTER TABLE Vehicle
ADD EngineSize DECIMAL(2,1) UNSIGNED CHECK(EngineSize <=7.9);

```

14.

The Song table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create a view named MyMusic that contains the Title, Genre, and Year columns for all songs. Ensure your result set returns the columns in the order indicated.

CREATE VIEW MyMusic AS

SELECT Title, Genre, Year

FROM Song;

15.

A database has a view named MyMusic.

Write a SQL statement to delete the view named MyMusic from the database.

DROP VIEW MyMusic;

16.

The Pet table has the following columns:

PetID – integer

Name - variable-length string

Breed - variable-length string

Birthdate - date

Write a SQL statement to modify the Pet table to make the PetID column the primary key.

ALTER TABLE Pet

ADD PRIMARY KEY (PetID);

17.

The Dog table has the following columns:

dogID - integer, primary key

name - variable-length string

breedID - integer

birthdate - date

The Breed table has the following columns:

breedID—integer

breedDescription—varchar

Releases—integer

Write a SQL statement to designate the breedID column in the Dog table as a foreign key to the breedID column in the Breed table.

ALTER TABLE Dog

ADD FOREIGN KEY (breedID) REFERENCES Breed (breedID);

18.

The Song table has the following columns:

ID—integer, primary key

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL statement to create an index named idx_year on the Year column of the Song table.

CREATE INDEX idx_year

ON Song (Year);

19.

The Podcast table has the following columns:

podcastID—integer, primary key, auto_increment

Title—variable-length string

Speaker—variable-length string

Genre – variable-length string

Minutes—integer

The following data needs to be added to the Podcast table:

Title	Speaker	Genre	Minutes
Rock Painting	Kecia McDonald	No Data	25

INSERT INTO Podcast (Title, Speaker, Genre, Minutes)
VALUES ('Rock Painting', 'Kecia McDonald', NULL, 25);

20.

The Podcast table has the following columns:

PodcastID—integer, primary key, auto_increment

Title—variable-length string

Speaker—variable-length string

Minutes—integer

Write a SQL statement to delete the row with the ID value of 33 from the Podcast table.

DELETE FROM Podcast
WHERE PodcastID = 33;

21.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Cost—positive decimal

Write a SQL query that will **add \$10 (for charity) to the cost of any book** that costs more than \$100 dollars.

UPDATE Book
SET Cost = Cost + 10
WHERE Cost > 100;

22. Which query illustrates performing an outer join of the Movie table with a different table?

- a. SELECT M.Title, A.Name
FROM Movie M, Actor A
WHERE M.ActorID = A.ActorID;
- b. SELECT M.Title, A.Name
FROM Movie M, Actor A
LEFT JOIN M.ActorID = A.MovieID;
- c. SELECT M.Title, A.Name
FROM Movie M
RIGHT JOIN Actor A
ON M.ActorID = A.ActorID; I spaced this out, to show you how I would view this on the OA.
- d. SELECT M.Title, A.Actor FROM Movie M
INNER JOIN Actor A ON M.ActorID = A.ActorID

This question is asking which statement illustrates an OUTER JOIN, not an INNER JOIN. The only 2 statements above that **include outer joins are b and c (LEFT JOIN and RIGHT JOIN).**

B is an invalid statement, so C has to be the answer. Please review the JOINS and Aggregate Functions PPT to get a good understanding of JOINS the OA way!

23.

Assume there are two tables, A and B.

Which rows will always be included in the result set if Table A is inner joined with Table B?

- a. Only rows in Tables A and B that share the join condition
- b. All rows in Table B
- c. All rows in Table A
- d. Only rows in Tables A and B that do not share the join condition.

24.

The database contains a table named Book.

Write a SQL query to return all data from the Book table without directly referencing any column names.

```
SELECT * FROM Book;
```

25.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to retrieve the Title and Genre values for all records in the Book table with a Year value of 2020. Ensure your result set returns the columns in the order indicated.

```
SELECT Title, Genre
```

```
FROM Book
```

```
WHERE Year = 2020;
```

26.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to display all Title values in alphabetical order A–Z.

This question is testing to see if the student knows how to use the ORDER BY clause. Here is the general syntax: ORDER BY clauses **SORT** or **DISPLAY** the data in a particular order.

```
SELECT column1, column2, ...  
FROM table_name  
ORDER BY column1, column2, ... ASC|DESC;
```

And here is the correct answer.

```
SELECT Title
```

```
FROM Book
```

```
ORDER BY Title ASC; On the OA, I would put ASC (even if the default is ASC).
```

OR

```
SELECT Title
```

```
FROM Book
```

```
ORDER BY Title; This gets full points in Zybooks because the default is ASC, but don't do it on the OA.
```

27.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to output the unique Genre values and the number of books with each genre value from the Book table as GenreCount. Sort the results by the Genre in alphabetical order A–Z. Ensure your result set returns the columns in the order indicated.

In this case you are being asked to count the number of Movies having each type of Genre. For this reason, you must use GROUP BY to group all moves with each Genre together and then do a count. The attribute you are grouping on must be listed first.

```
SELECT DISTINCT Genre, COUNT(*) AS GenreCount  
FROM Book  
GROUP BY Genre  
ORDER BY Genre ASC;
```

Please take a look at these resources:

[GROUP BY](#)
[COUNT](#)

28.

The **Book** table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

The **YearSales** table has the following columns:

Year—integer

TotalSales—bigint unsigned

Releases—integer

Write a SQL query to display both the Title(if available) and the TotalSales for all YearSales.

Ensure your result set returns the columns in the order indicated.

Here, you must have a JOIN because you are pulling a value from each of the two tables. The stipulation means it has to be an outer join (left or right). It is a RIGHT join because you want ALL YearSales.

```
SELECT Book.Title, YearSales.TotalSales
```

```
FROM Book
```

```
RIGHT JOIN YearSales
```

```
ON Book.Year = YearSales.Year;
```

29.

The Book table has the following columns:

ID—integer, primary key, auto_increment

Title—variable-length string

Genre—variable-length string

Year—integer

Write a SQL query to return how many books have a Year value of 2019.

```
SELECT COUNT(*)
```

```
FROM Book
```

```
WHERE Year = 2019;
```

30. A(n) _____ is a query that is embedded (or nested) inside another query.

- a. alias
- b. operator
- c. subquery
- d. view

31. **All changes to a table structure** are made using the _____ command, followed by a keyword that produces the specific changes a user wants to make.

- a. ALTER TABLE
- b. UPDATE TABLE
- c. COMMIT TABLE
- d. DELETE TABLE

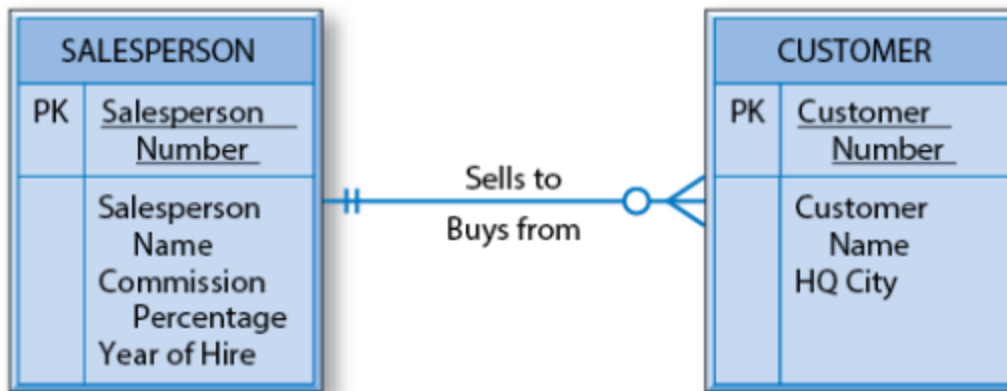
32. The SQL aggregate function that gives the number of rows containing non-null values for a given column is _____.

- a. COUNT
- b. MIN
- c. MAX
- d. SUM

33. The **ON** condition of a JOIN is generally composed of an equality comparison between the foreign key and the primary key of related tables.

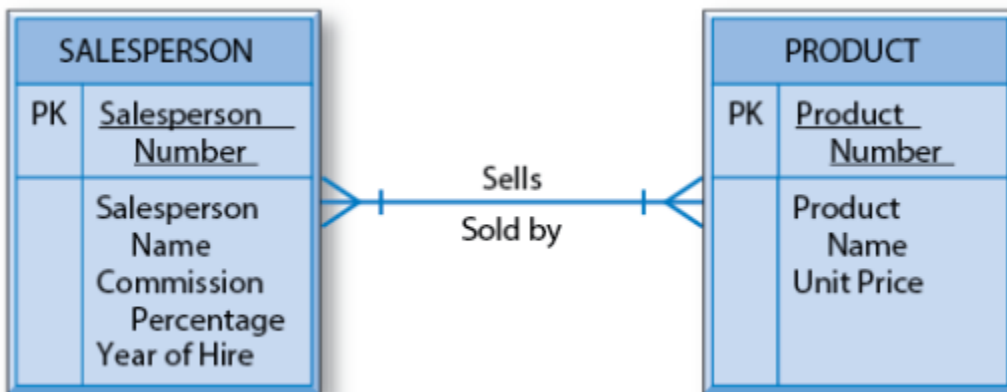
34. Which kind of relationship is displayed in the entity-relationship diagram below?

- a. Binary one-to-one
- b. Unary many-to-many
- c. Ternary one-to-one
- d. Binary one-to-many
- e. Unary one-to-one



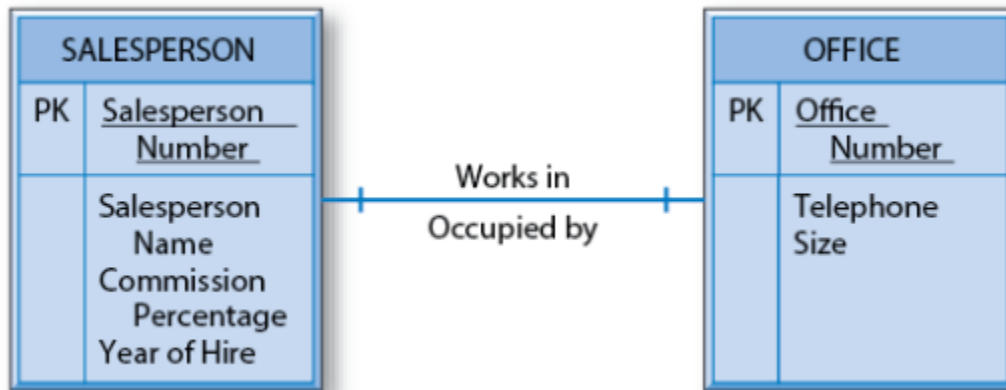
35. Which kind of relationship is displayed in the entity-relationship diagram below?

- a. Binary one-to-one
- b. Unary many-to-many
- c. Ternary one-to-one
- d. Binary many-to-many
- e. Unary one-to-one



36. Which kind of relationship is displayed in the entity-relationship diagram below?

- a. Binary one-to-one
- b. Unary many-to-many
- c. Ternary one-to-one
- d. Binary many-to-many
- e. Unary one-to-one



37. Which data type can be designated to allow for storage of dates

- a. VARCHAR
- b. INTEGER
- c. DECIMAL
- d. **TIMESTAMP/DATE**

38.

Patient

patient_id	first_name	last_name	birthdate
101	Amy	Lin	1986-04-02
102	Viya	Prasad	1999-03-26
104	Blanca	Garcia	2000-12-11
107	Shelia	Brown	2004-11-11

Exam

exam_id	exam_date	exam_reason	patient_id
1	101	Wellness check	104
2	102	vaccine	107

Which query would produce a result set that listed all of the patients, **regardless of whether they had an appointment** in the Exam table or not.

- a. SELECT patient_id, first_name, last_name, exam_date
FROM Patient RIGHT JOIN Exam

ON Patient.patient_id = Exam.patient_id;

b. SELECT patient_id, first_name, last_name, birthdate, exam_date
FROM Patient LEFT JOIN Exam
ON Patient.patient_id = Exam.patient_id;

c. SELECT patient_id, first_name, last_name, birthdate, exam_date
FROM Patient INNER JOIN Exam
ON Patient.patient_id = Exam.exam_id;

d. SELECT patient_id, first_name, last_name, birthdate, exam_date
FROM Patient JOIN Exam
ON Patient.patient_id = Exam.patient_id;

39.

Models			
<u>model_id</u>	<u>lname</u>	<u>birthdate</u>	<u>model_type_id</u>
101	Almquist	9/1/05	CM
102	Chang	12/30/85	CP
103	Prasad	5/25/80	HF
104	Greene	5/23/83	PS

Model_Type	
<u>model_type_id</u>	<u>hourly_fee</u>
CM	500.00
HF	800.00
CP	800.00
PS	700.00
PM	700.00

Which query would produce a result set that listed all of the Model Types, regardless of whether a model was assigned to that type.

a. SELECT model_id, lname, Models.model_type_id, hourly_fee
FROM Models
RIGHT JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;

- b. `SELECT model_id, lname, Models.model_type_id, hourly_fee
FROM Models
LEFT JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;`
- c. `SELECT model_id, lname, Models.model_type_id, hourly_fee
FROM Models
INNER JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;`
- d. `SELECT model_id, lname, Models.model_type_id, hourly_fee
FROM Models
JOIN Model_Type
ON Models.model_type_id = Model_Type.model_type_id;`

40.

Using Q39 tables, write a query to list all model IDs, the model type they are assigned and their hourly fees.

```
SELECT Models.model_id, Models.model_type_id, Model_Type.hourly_fee  
FROM Models  
INNER JOIN Model_Type  
ON Models.model_type_id = Model_Type.model_type_id;
```

41.

Write a query to delete Amy Lin from the Customer table shown below.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip	Mobile Phone
101	Amy	Lin	123 A Street	Seattle	WA	98118	555-555-5555
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118	555-555-5556
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212	555-887-9264
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217	555-209-2784

DELETE FROM Customer
WHERE CustomerID = 101;

42. Write a SQL statement to update Blanca Garcia's phone number to 555-222-1234.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip	Mobile Phone
101	Amy	Lin	123 A Street	Seattle	WA	98118	555-555-5555
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118	555-555-5556
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212	555-887-9264
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217	555-209-2784

UPDATE Customer
SET MobilePhone = '555-222-1234'
WHERE CustomerID = 104;

43. Write a SQL statement to retrieve all of the Customers from Seattle from the table below.

Customer

Customer ID	First Name	Last Name	Address	City	State	Zip	Mobile Phone
101	Amy	Lin	123 A Street	Seattle	WA	98118	555-555-5555
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118	555-555-5556
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212	555-887-9264
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217	555-209-2784

SELECT *
FROM Customer
WHERE City = 'Seattle';

44. Write the CREATE TABLE statement to create the Exam table below with exam_id and patient_id as integers, and exam reason to have 250 characters of variable length. Make sure to designate the primary key and foreign key of the table.

Patient

<u>PK patient_id</u>
first name
last name
birthdate

Exam

<u>PK exam_id</u>
exam date
exam reason
FK patient_id

```
CREATE TABLE Exam (
    exam_id      INT,
    exam_date    DATE,
    exam_reason  VARCHAR(250),
    patient_id   INT,
    PRIMARY KEY (exam_id),
    FOREIGN KEY (patient_id) REFERENCES Patient(patient_id)
);
```

45. Write a query to pull all of the data from the Bagel table below. Order the result set alphabetically by Bagel ID.

BagelID	BagelName	BagelDescription	Cost/Bagel
RB	Radically Raisin Bagel	Raisin Bagel	2.00
EB	All the Stuff Bagel	Everything Bagel	2.35
CB	Chip off the Block Bagel	Chocolate Chip Bagel	2.25
MB	Keepin' it Healthy Bagel	Multigrain Bagel	2.00
NB	Classic New York Bagel	New York Bagel	1.75

```
SELECT *
FROM Bagel
ORDER BY BagelID ASC;
```

46. Write a query to count the number of customers in each city.

Customer ID	First Name	Last Name	Address	City	State	Zip
101	Amy	Lin	123 A Street	Seattle	WA	98118
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217
108	Anna	Cruz	444 Apple Lane	Troutdale	OR	97708
110	Amy	Smith	39878 Acorn Place	Chicago	IL	60435

SELECT City, COUNT(*)

FROM Customer

GROUP BY City;

47. Write a query that will return all of the cities from the Customer table above. Do not include duplicates -list each city only once.

SELECT DISTINCT City

FROM Customer;

48. Write a query to list customers from Chicago or Seattle.

Customer ID	First Name	Last Name	Address	City	State	Zip
101	Amy	Lin	123 A Street	Seattle	WA	98118
102	Viya	Prasad	111 B Avenue	Seattle	WA	98118
104	Blanca	Garcia	8674 Main Street	Portland	OR	97212
107	Shelia	Brown	22815 Foster Road	Portland	OR	97217
108	Anna	Cruz	444 Apple Lane	Troutdale	OR	97708
110	Amy	Smith	39878 Acorn Place	Chicago	IL	60435

```
SELECT *  
FROM Customer  
WHERE City = 'Chicago' OR City = 'Seattle';
```

49.

student_grades

studentID	lastName	grade
44444	Jones	B
24352	Chang	A
98724	Wu	B
98789	Chatham	A
98887	Fredricks	C
00878	Moore	A
22078	Prasad	C
79692	Perez	A

Refer to the given SELECT statement and the table above.

```
SELECT grade, COUNT(*) AS gradeCount
```

FROM student_grades

GROUP BY grade;

Which clause added to the statement finds grades with a count greater than 2?

a. WITH gradeCount <> DUPLICATE

b. NOT gradeCount = NULL

c. **HAVING gradeCount > 2**

d. INCLUDING gradeCount > 2

50.

Using the student_grades table listed above, write a query to count **how many students** have **each grade** (A, B, and C).

SELECT grade, COUNT(*)

FROM student_grades

GROUP BY grade;

51. Using the Model_Types table in the E-R diagram above, write a query to find the highest model hourly fee.

SELECT MAX(hourly_fee)

FROM Model_Types;

52. Write a query to **change the hourly_fee to add \$100 more** for model types with an **hourly_fee less than 600.00**.

UPDATE Model_Types

SET hourly_fee = hourly_fee + 100

WHERE hourly_fee < 600;

53.

Which query illustrates performing an outer join of the Movie table with a different table?

a. SELECT M. Title, A.Actor FROM Movie M, Actor A
WHERE M.ActorID = A.ID

b. SELECT M.Title, A.Actor FROM Movie M
LEFT JOIN Movie MB ON M.ID = MB.ID, Actor A

c. SELECT M. Title, A.Actor FROM Movie M
INNER JOIN Actor A ON M. ActorID = A.ID

d. SELECT M. Title, A.Actor FROM Movie M
RIGHT JOIN Actor A ON M.ActorID = A.ID

54. Which rows will always be included in the result set if Table A is right joined with Table B?

- a. Only rows in Tables A and B that share the join condition
- b. All rows in Table B and any rows in Table A that share the join condition
- c. All rows in Table A and any rows in Table B that share the join condition
- d. Only rows in Tables A and B that do not share the join condition

55. Canadians like long movie titles (Bill and Ted's Excellent Adventures). Modify the **MovieName** column in the Movie table so that the variable length string has a **maximum of 50 characters**.

ALTER TABLE Movie

CHANGE MovieName MovieName VARCHAR(50);

This question is different from a question that asks you to change the name of the column and the number of characters. If it said **RENAME** the column to **CanadianMovieName** and change the length to a maximum of 50 characters, the correct answer would be:

ALTER TABLE Movie

CHANGE MovieName CanadianMovieName VARCHAR(50);

56.

Write a SQL statement to pull the **EmployeeType** and the **lowest salary by EmployeeType** in the Employee table:

```
SELECT EmployeeType, MIN(Salary)
FROM Employee
GROUP BY EmployeeType;
```

Here is what the result could look like:

Employee Type	MIN(Salary)
Associate	40000
Supervisor	60000
Manager	80000
Director	100000